$$\frac{}{\Delta \mid \Gamma\,;\, x : A \vdash x : A}\ \mathsf{hyp}$$

$$\frac{}{\Delta \mid \Gamma\,;\, \cdot \vdash () : \mathbf{1}}\ \mathbf{1}I \qquad \frac{\Delta \mid \Gamma\,;\, \Omega \vdash e : \mathbf{1} \quad \Delta \mid \Gamma\,;\, \Omega_L \Omega_R \vdash e' : C}{\Delta \mid \Gamma\,;\, \Omega_L \Omega \Omega_R \vdash \mathbf{match}\ e\ (() \Rightarrow e') : A}\ \mathbf{1}E$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega\,(x : A) \vdash e : B}{\Delta \mid \Gamma\,;\, \Omega \vdash \lambda x.\, e : A \twoheadrightarrow B}\ {\twoheadrightarrow}I \qquad \frac{\Delta \mid \Gamma\,;\, \Omega \vdash e_1 : A \twoheadrightarrow B \quad \Delta \mid \Gamma\,;\, \Omega_A \vdash e_2 : A}{\Delta \mid \Gamma\,;\, \Omega\,\Omega_A \vdash e_1\, e_2 : B}\ {\twoheadrightarrow}E$$

$$\frac{\Delta \mid \Gamma\,;\, (x : A)\,\Omega \vdash e : B}{\Delta \mid \Gamma\,;\, \Omega \vdash \lambda x.\, e : A \rightarrowtail B}\ {\rightarrowtail}I \qquad \frac{\Delta \mid \Gamma\,;\, \Omega \vdash e_1 : A \rightarrowtail B \quad \Delta \mid \Gamma\,;\, \Omega_A \vdash e_2 : A}{\Delta \mid \Gamma\,;\, \Omega_A\,\Omega \vdash e_1\, e_2 : B}\ {\rightarrowtail}E$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega_A \vdash e_1 : A \quad \Delta \mid \Gamma\,;\, \Omega_B \vdash e_2 : B}{\Delta \mid \Gamma\,;\, \Omega_A\,\Omega_B \vdash (e_1, e_2) : A \bullet B}\ \bullet I$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega \vdash e : A \bullet B \quad \Delta \mid \Gamma\,;\, \Omega_L\,(x : A)\,(y : B)\,\Omega_R \vdash e' : C}{\Delta \mid \Gamma\,;\, \Omega_L\,\Omega\,\Omega_R \vdash \mathbf{match}\ e\ ((x, y) \Rightarrow e') : C}\ \bullet E$$

$$\frac{\Delta, \alpha\ \mathsf{type} \mid \Gamma\,;\, \Omega \vdash e : A}{\Delta \mid \Gamma\,;\, \Omega \vdash e : \forall \alpha.\, A}\ \forall I \qquad \frac{\Delta \mid \Gamma\,;\, \Omega \vdash e : \forall \alpha.A(\alpha) \quad \Delta \vdash B\ \mathsf{type}}{\Delta \mid \Gamma\,;\, \Omega \vdash e : A(B)}\ \forall E$$

$$\frac{}{\Delta \mid \Gamma, x : A\,;\, \cdot \vdash x : A}\ \mathsf{hyp}$$

$$\frac{\Delta \mid \Gamma, x : A\,;\, \Omega \vdash e : B}{\Delta \mid \Gamma\,;\, \Omega \vdash \lambda x.\, e : A \to B}\ {\to}I \qquad \frac{\Delta \mid \Gamma\,;\, \Omega \vdash e_1 : A \to B \quad \Delta \mid \Gamma\,;\, \cdot \vdash e_2 : A}{\Delta \mid \Gamma\,;\, \Omega \vdash e_1\, e_2 : B}\ {\to}E$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega_A \vdash e_1 : A \quad \Delta \mid \Gamma\,;\, \Omega_B \vdash e_2 : B}{\Delta \mid \Gamma\,;\, \Omega_B\,\Omega_A \vdash (e_1, e_2) : A \circ B}\ \circ I$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega \vdash e : A \circ B \quad \Delta \mid \Gamma\,;\, \Omega_L\,(y : B)\,(x : A)\,\Omega_R \vdash e' : C}{\Delta \mid \Gamma\,;\, \Omega_L\,\Omega\,\Omega_R \vdash \mathbf{match}\ e\ ((x, y) \Rightarrow e') : C}\ \circ E$$

$$\frac{(k \in L) \quad \Delta \mid \Gamma\,;\, \Omega \vdash e : A_k}{\Delta \mid \Gamma\,;\, \Omega \vdash k(e) : \oplus\{\ell : A_\ell\}_{\ell \in L}}\ \oplus I$$

$$\frac{\Delta \mid \Gamma\,;\, \Omega \vdash e : \oplus\{\ell : A_\ell\}_{\ell \in L} \quad (\Delta \mid \Gamma\,;\, \Omega_L\,(x_\ell : A_\ell)\,\Omega_R \vdash e_\ell : A_\ell) \quad (\forall \ell \in L)}{\Delta \mid \Gamma\,;\, \Omega_L\,\Omega\,\Omega_R \vdash \mathbf{match}\ e\ \{\ell(x_\ell) \Rightarrow e_\ell\}_{\ell \in L} : C}\ \oplus E$$